

ZAMEEN360°

3D Visualization buying and selling



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CHAPTER 1

1. Introduction

Purpose

Zameen360 is a next-generation real estate platform designed to solve the inefficiencies of traditional property portals. While traditional systems often suffer from mismatched listings and high volumes of unverified data, Zameen360 integrates **Artificial Intelligence**, **3D Visualization**, and **Behavioral Risk Scoring** to create a high-trust environment for buyers and sellers.

Scope

It includes the development of a **React-based web interface** and a **Node.js backend** capable of processing automated **NIC identity verification** and **semantic property searches**. The boundary of the project encompasses the integration of **3D floor plan generation** from 2D images and the implementation of a **Risk Scoring Engine** to flag anomalous pricing or suspicious user behavior. Geographically and data-wise, the initial scope is limited to residential and commercial listings within **Pakistan**, utilizing integrations with the **Punjab Land Records Authority (PLRA)** for document cross-referencing.

Definitions, Acronyms, and Abbreviations

- **PLRA:** Punjab Land Records Authority
- **CRUD:** Create, Read, Update, Delete
- **JWT:** JSON Web Token
- **UML:** Unified Modeling Language
- **GDPR:** General Data Protection Regulation
- **SRS:** Software Requirements Specification

References

- IEEE Standard for Software Test Documentation (829-2008)
- Regulatory Guidelines (e.g., GDPR)

Overview

This document provides a comprehensive overview of the **Zameen360** requirements. It includes detailed functional descriptions, performance benchmarks, security measures, and critical system attributes designed to facilitate a high-trust real estate ecosystem. As a living document, this SRS will be updated throughout the project lifecycle to reflect evolving market needs and technical refinements.

Background

Traditional real estate portals in Pakistan often suffer from a lack of transparency, leading to mismatched expectations and a high volume of unverified or "fake" listings. By integrating **Artificial Intelligence (AI)**, **3D Visualization**, and **Behavioral Risk Scoring**, Zameen360 aims to reduce transaction friction, improve property matching quality, and provide actionable intelligence to both buyers and sellers.

Problem Statement

Property seekers struggle to find verified listings that precisely match their preferences, while legitimate sellers find their posts buried under a high volume of unverified data. This creates significant information asymmetry and trust issues in the recruitment of property assets. There is a critical need for an intelligent system that bridges this gap and continuously learns from user interactions to improve the reliability of the marketplace.

Operating Environment

- **Client Side:** Compatible with all major browsers (Chrome, Firefox, Safari, Edge) on desktops, tablets, and smartphones.
- **Server Side:** Implemented using a scalable backend framework (e.g., Node.js,) hosted in a cloud environment (e.g., AWS).
- **Database:** Utilizes robust postgres SQL solutions to ensure data integrity and fast access

2. System Features

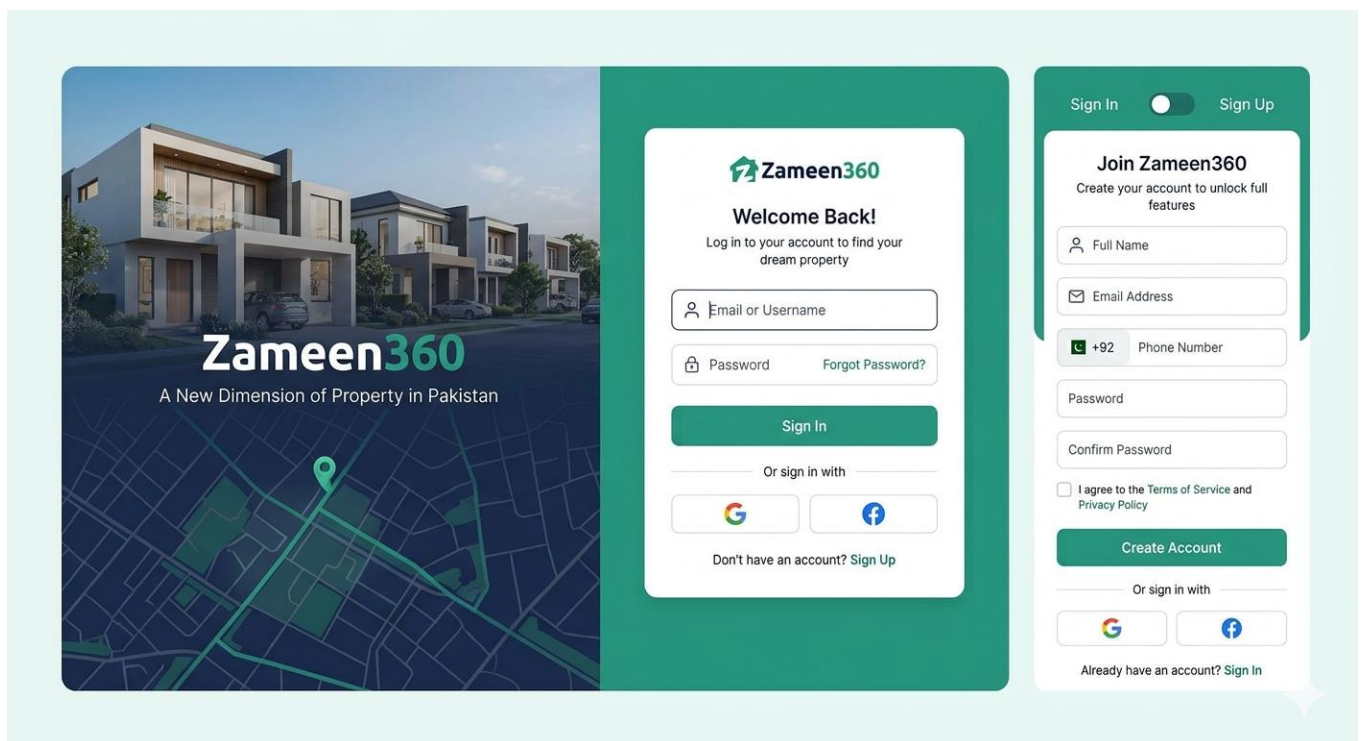
User Registration and Login

Description:

Secure user account creation, login, and password recovery for job seekers and employers through email authentication and secure credentials.

Details:

- Registration forms capture necessary user information.
- Two-factor authentication for enhanced security.
- Role-based access control distinguishes job seeker, employer, and admin functionalities.



2.4 User Profile Module

Description:

- Owns all user data, preferences, and history. Acts as the user's permanent record across the platform — from saved searches to verification badges.
- RESPONSIBILITIES
- Basic profile: name, contact, location
- Verification status and tier badge display
- Search history and saved properties
- Notification preferences
- Transaction history overview
- Trust tier display (Basic → Verified → Trusted → Professional)

Listings Module

Description:

- Manages the full lifecycle of a property listing — from draft to sold. A 'property' is separate from a 'listing': one property can have multiple listings over time.

Responsibilities

- Create / Edit / Delete / Pause listings
- Types: Residential, Commercial, Plot, File
- Rich media: photos, video, virtual tour
- Status machine: Draft → Under Review → Active → Under Contract → Sold/Rented
- Duplicate detection (same property, different seller = fraud flag)
- Auto-expiry for stale listings
- Full price change history log

Chapter 2

Functional Requirements-Core Platform Features

User Management & Profile Requirements

Requirement Identifier: REQ-001

Requirement Title: User Account Registration

Requirement Description: The system shall allow users to create an account by providing an email address, a password, and their full name.

Rationale: To enable users to access personalized features and track their property activities.

Acceptance Criteria: All fields must be validated, and a confirmation email must be sent upon successful registration.

Source: Stakeholder Interview

Priority: Must Have

Dependencies: None

Requirement Identifier: REQ-002

Requirement Title: AI-Driven Identity Verification

Requirement Description: The system shall use AI to extract data from an uploaded NIC and verify it against NADRA standards.

Rationale: To ensure user legitimacy and reduce the prevalence of fake listings in the marketplace.

Acceptance Criteria: The system must accurately extract the Name and NIC number; verification status must update in the user profile.

Source: AI Verification Strategy

Priority: Must Have

Dependencies: REQ-001

Requirement Identifier: **REQ-003**

Requirement Title: **3D Media & Tour Upload**

Requirement Description: The system shall allow users to upload 360-degree panoramas and automatically generate 3D floor plans.

Rationale: To provide immersive virtual tours and increase buyer confidence without physical site visits.

Acceptance Criteria: 3D floor plans must be generated and viewable within the listing page after processing.

Source: User Stories

Priority: Must Have

Dependencies: REQ-006

Requirement Identifier: **REQ-04**

Requirement Title: **Semantic AI Search**

Requirement Description: The system shall provide an AI-based search function that understands natural language queries for property discovery.

Rationale: To allow users to find properties using intuitive phrases rather than just rigid filters.

Acceptance Criteria: Querying "3 bed near a park" must return properties with those specific attributes ranked by relevance.

Source: AI Search Strategy

Priority: Must Have

Dependencies: REQ-006

Requirement Identifier: **REQ-021**

Requirement Title: **Automated 3D Floor Plan Generation**

Requirement Description: The system shall process uploaded 2D architectural sketches or high-angle room photos to generate a navigable 3D floor plan model.

Rationale: To provide users with a spatial understanding of the property layout that standard photography cannot convey.

3. Non-Functional Requirements

Performance Requirements

- **Response Times:** The system must return search results within 2 seconds.
- **AI Matching:** AI-driven recommendations should be generated within 3 seconds of profile/job updates.
- **Load Capacity:** Capable of handling up to 1 million concurrent users during peak times.

Security Requirements

- **Data Protection:** All personal data must be encrypted in transit (TLS) and at rest (AES-256).
- **Authentication:** Two-factor authentication for employer accounts, with periodic security audits.
- **Compliance:** Adheres to GDPR, CCPA, and other relevant regulations.

Reliability

- **Uptime:** System availability shall be 99.9%, ensuring minimal service interruptions.
- **Fault Tolerance:** Failover mechanisms and regular disaster recovery drills must be implemented.

Availability

- **24/7 Access:** The platform must be accessible 24 hours a day, 7 days a week, with scheduled maintenance windows announced in advance.

Maintainability and Supportability

- **Modular Architecture:** Code shall be organized in well-defined modules to facilitate updates and debugging.
- **Documentation:** Comprehensive developer and user documentation will be maintained.

Usability

- **User Interface:** The design must be intuitive, following WCAG 2.1 guidelines for accessibility.
- **Onboarding:** New users receive guided tutorials and contextual help throughout their experience.
- **Feedback Mechanisms:** Built-in feedback options to continuously improve the UX.

Scalability

- **Horizontal Scalability:** The system must support scaling out via cloud services (e.g., AWS, Azure) to manage increased loads without degradation.
- **Vertical Scalability:** Ability to upgrade system hardware (CPU, RAM) if necessary.
- **Database Scalability:** Supports sharding, replication, and caching mechanisms to handle large volumes of data and queries.

Portability

- **Cross-Platform Compatibility:** The application must run consistently across all major browsers (Chrome, Firefox, Safari, Edge) and display properly on mobile, tablet, and desktop devices.
- **Responsive Design:** The UI adapts seamlessly to a variety of screen sizes and resolutions.

4. Other Requirements

Legal and Regulatory Requirements

- **Data Privacy:** Comply with GDPR, CCPA, and local data protection laws.
- **Employment Laws:** Adhere to Equal Employment Opportunity regulations and non-discrimination policies.
- **Terms and Policies:** Clearly defined Terms of Service and Privacy Policy visible to all users.

Chapter 3

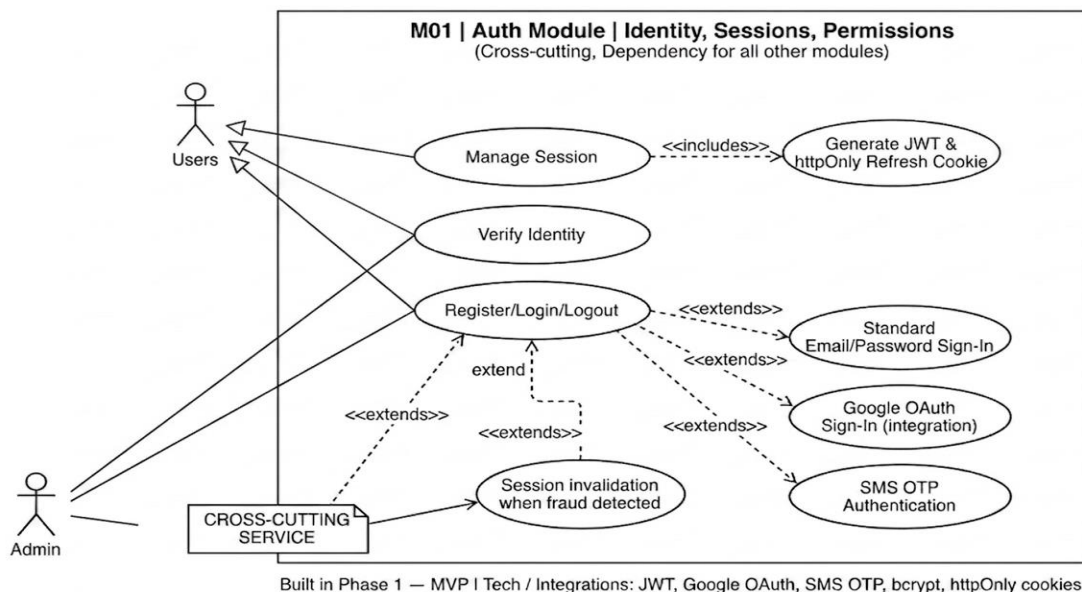
UML Use Case

UC-01-SignUp/Registration

Title	Sign Up
Actor	• User (Job Seeker) • Third-Party Authentication System (Google/Facebook)
Goal	Successfully create a new account
Precondition	User does not have an existing account
Main Flow	1. User provides email, password, first and last names. 2. System validates fields. 3. System sends a confirmation email. 4. User confirms email. 5. Account created successfully. - If email already exists: show error "Account already exists."
Alternative Flow	- If invalid email: prompt user to correct email.

Requirement

Linked REQ-001



UC-04-User Profile Management

Use Case: User Profile Management

Title: User Profile Management

Actor: User (Property Seeker or Seller)

Goal: Successfully view and edit personal profile information and account preferences.

Precondition: The user must be logged into the Zameen360 platform with an active, authenticated account.

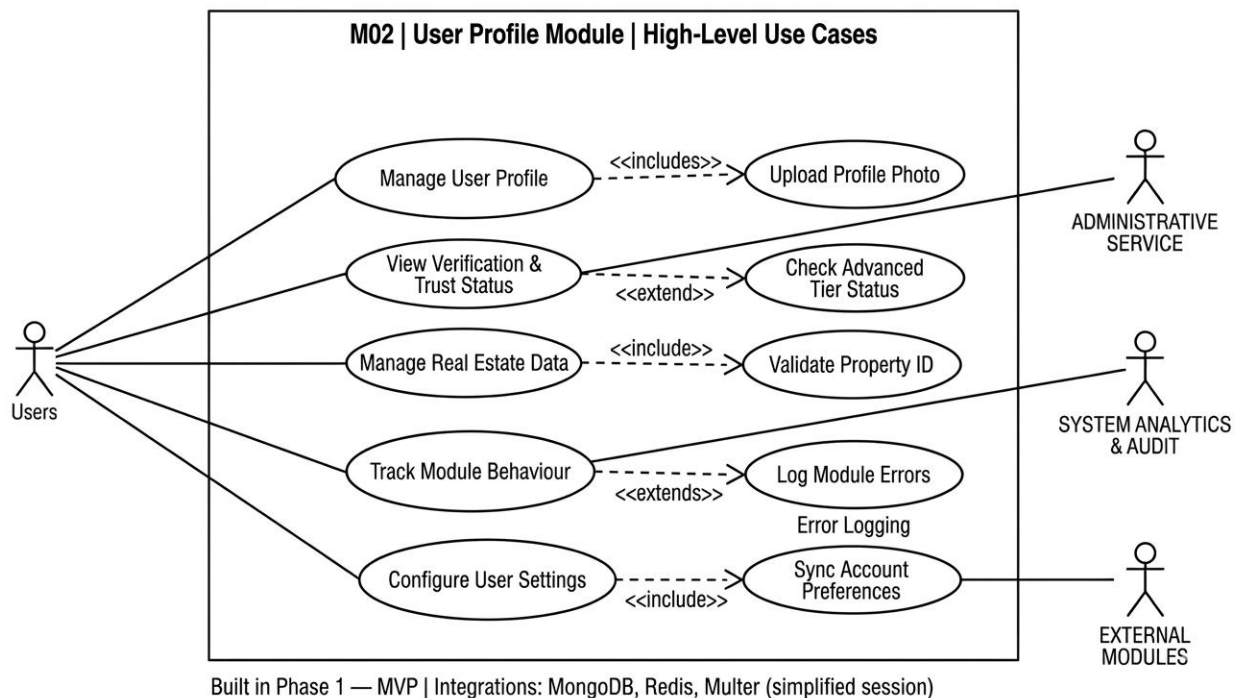
Main Flow:

1. The user navigates to the "Account Settings" or "Profile" section of the dashboard.
2. The system retrieves and displays the current profile information, such as Name, Contact Details, and NIC Verification Status.
3. The user updates specific details, which may include their phone number, profile picture, or preferred property categories.
4. The user clicks the "Save Changes" button.
5. The system validates the new data for correct formatting and updates the database.
6. The system displays a confirmation message indicating the profile has been successfully updated.

Alternative Flow:

- **Invalid Input:** If the user enters an incorrectly formatted phone number or an unsupported image type, the system shall show a validation error and prompt for correction.
- **Network/Server Error:** If the connection is lost during the update, the system shall display a message stating "Unable to save changes. Please check your connection and try again later".

Requirement Linked: REQ-004



Use Case: Property Listing Management

Title: Property Listing Management

Actor: User (Seller or Real Estate Agent)

Goal: Successfully create, edit, and manage the lifecycle of a real estate listing.

Precondition: User must be logged in and have completed the AI-driven identity verification process.

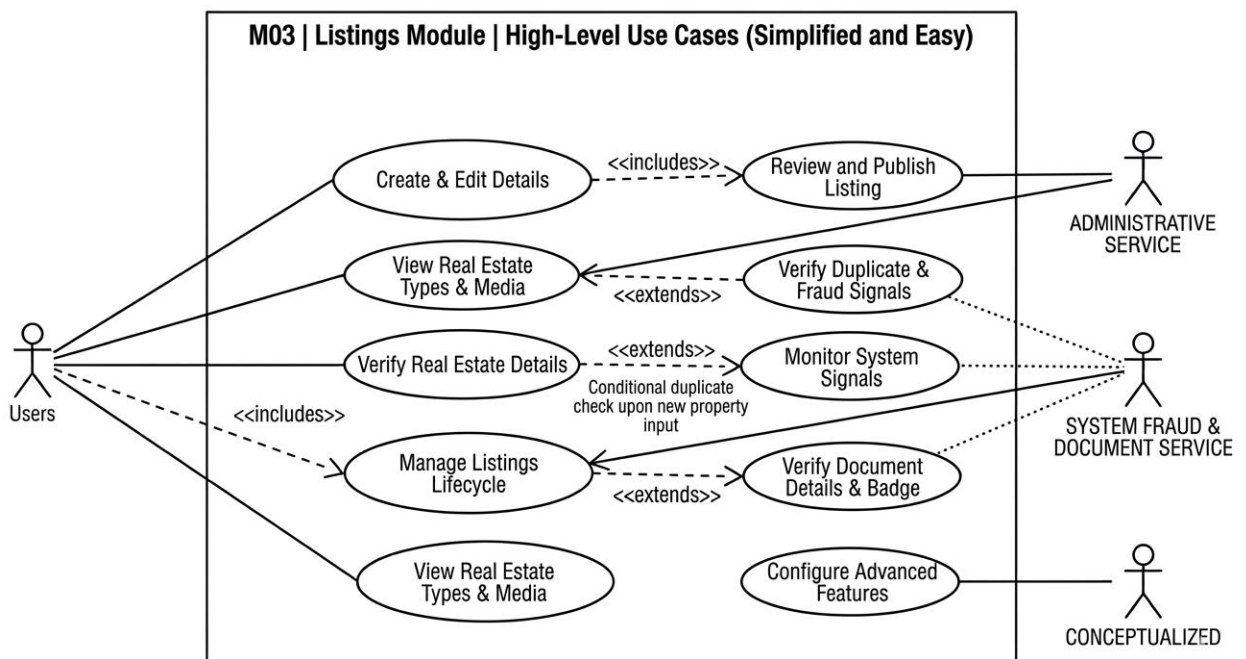
Main Flow:

1. The user navigates to the "Manage Listings" section and selects "Create New Listing".
2. The user inputs property details, including type, price, and location coordinates.
3. The user uploads property media, such as high-resolution images or 360-degree panoramas.
4. The system automatically triggers a background check for duplicate listings and potential fraud signals.
5. The user submits the listing for review.
6. The Administrative Service reviews the details and publishes the listing to the live marketplace.
7. The system displays a confirmation that the listing is now "Active".

Alternative Flow:

- **Duplicate/Fraud Detected:** If the system identifies a matching existing property or suspicious data, it flags the listing for the System Fraud & Document Service to verify document details before publishing.
- **Draft Saving:** If the user is unable to complete the listing, they can save it as a draft to resume the lifecycle management at a later time.

Requirement Linked: REQ-006, REQ-018



M03 — Phase 1 MVP | Simplified for Clarity. For full detail, see comprehensive documentation.

Use Case: 3D Model & Floor Plan Generation

Title: 3D Model & Floor Plan Generation

Actor: User (Seller or Real Estate Agent)

Goal: To automatically generate a navigable 3D model and interactive floor plan from uploaded 2D media.

Precondition: User must have an active listing draft and a high-speed internet connection for processing large media files.

Main Flow:

1. The user selects the "Generate 3D Model" option within the listing editor.
2. The system prompts the user to upload high-angle room photos or architectural 2D sketches.
3. The user uploads the files and clicks "Process Media".
4. The system utilizes AI computer vision to analyze spatial dimensions and depth.
5. The system renders a navigable 3D floor plan and stitches 360-degree views where applicable.
6. The user reviews the generated model in the "Preview" window.
7. The user clicks "Attach to Listing" to make the 3D tour visible to potential buyers.

Alternative Flow:

- Low Image Quality: If the uploaded photos are too dark or blurry for the AI to process, the system shall trigger a validation error: "Image quality insufficient for 3D rendering. Please provide clearer photos".
- Processing Timeout: If the server is overloaded, the system shall show a "Processing in Progress" status and notify the user via the dashboard once the model is ready for review.

Requirement Linked: REQ-021, REQ-022, REQ-025

ERD DIAGRAM

ERD Diagram – Real Estate Website

